

High Fidelity FM Transmitter

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21st August 2025

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1 Aim

The primary objective of this experiment is to simulate and analyze a high-fidelity FM transmitter circuit to understand:

- The working principles of PLL-based frequency synthesis
- Audio signal processing and frequency modulation techniques
- Circuit performance parameters including frequency stability and modulation characteristics
- Power amplification and antenna coupling mechanisms
- Compliance with broadcasting standards and regulations

2 Theory

2.1 Frequency Modulation Fundamentals

Frequency Modulation (FM) is a method of encoding information in a carrier wave by varying the instantaneous frequency of the wave. The mathematical representation of an FM signal is:

$$s_{FM}(t) = A_c \cos \left(2\pi f_c t + 2\pi k_f \int_0^t m(\tau) d\tau \right) \quad (1)$$

where:

- A_c = carrier amplitude
- f_c = carrier frequency
- k_f = frequency deviation constant
- $m(t)$ = modulating signal (audio)

2.2 Phase-Locked Loop (PLL) Theory

A PLL is a control system that generates an output signal whose phase is related to the phase of an input signal. The basic PLL consists of:

- Phase detector/comparator
- Loop filter (low-pass filter)
- Voltage-controlled oscillator (VCO)
- Optional frequency divider for frequency synthesis

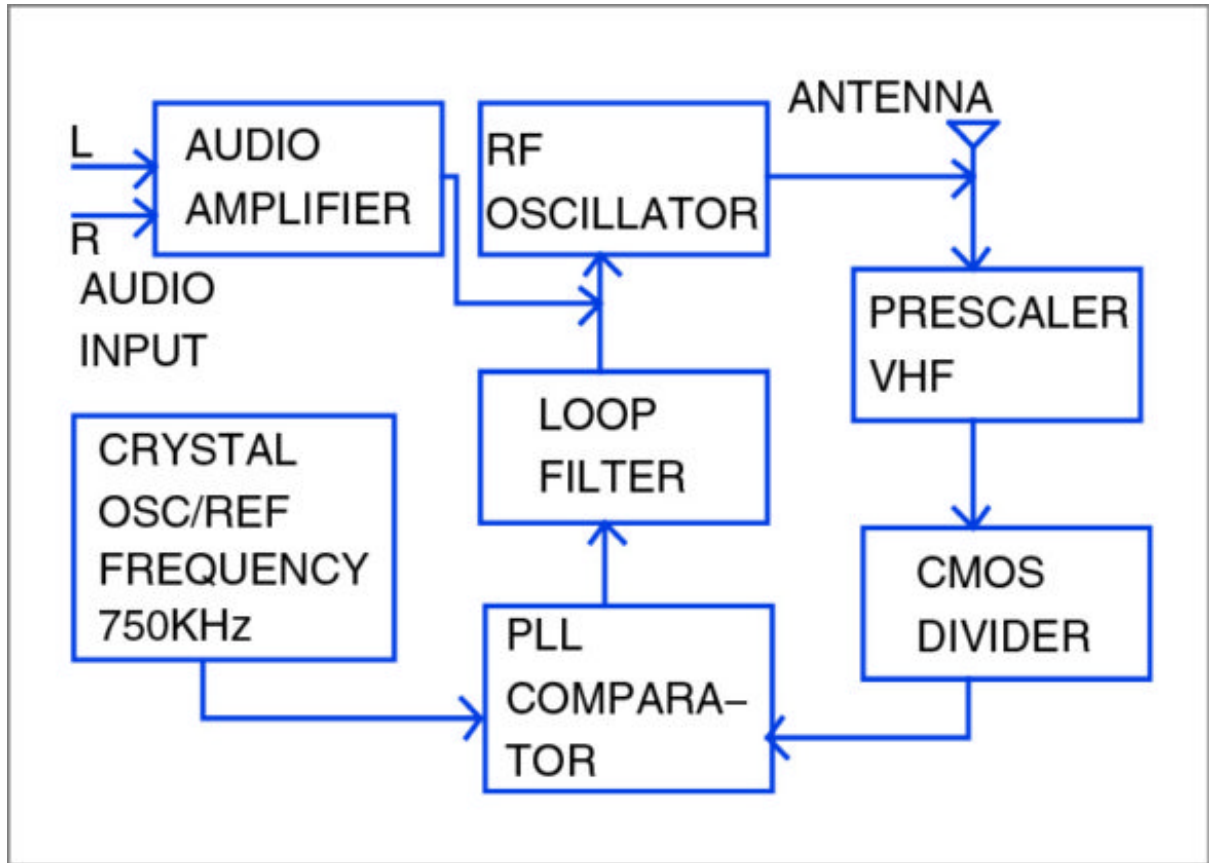


Figure 1: Block Diagram

The loop maintains lock when:

$$f_{out} = \frac{N \times f_{ref}}{M} \quad (2)$$

where N and M are the division ratios in the feedback and reference paths.

2.3 VHF Band Characteristics

The VHF (Very High Frequency) band spans 30-300 MHz, with FM broadcasting allocated to 88-108 MHz. Key characteristics include:

- Line-of-sight propagation
- Limited atmospheric interference
- High-quality audio transmission capability
- Requirement for precise frequency control

3 Circuit Description and Working

3.1 Block Diagram Analysis

The FM transmitter consists of several key functional blocks:

3.1.1 Audio Processing Stage

- **Audio Input:** Accepts left (L) and right (R) channel inputs
- **Audio Amplifier:** Provides necessary gain and impedance matching
- **Pre-emphasis:** Enhances high-frequency components for improved SNR

3.1.2 Frequency Generation and Modulation

- **Crystal Oscillator:** Provides 750 kHz reference frequency for stability
- **PLL Comparator:** Compares reference and feedback frequencies
- **Loop Filter:** Removes high-frequency components from phase detector output
- **RF Oscillator:** VCO generates carrier frequency, modulated by audio signal

3.1.3 Output Stage

- **Prescaler:** Divides VHF frequency for PLL feedback
- **CMOS Divider:** Further frequency division for phase comparison
- **Antenna:** Radiates the modulated RF signal

3.2 Detailed Circuit Analysis

3.2.1 Audio Amplification (IC6 - LM386)

The LM386 provides low-power audio amplification with:

- Voltage gain: 20-200 (depending on external components)
- Input impedance: 50k Ω
- Output power: Up to 1W into 8 Ω load
- Single supply operation: 4-12V

The gain is set by components between pins 1 and 8:

$$A_v = 20 \times \left(1 + \frac{R_{1-8}}{1.35k\Omega} \right) \quad (3)$$

3.2.2 Voltage Regulator (IC2 - 7805)

Provides stable +5V supply for digital circuits:

- Input voltage range: 7-25V
- Output voltage: 5V $\pm$ 4%
- Maximum output current: 1A
- Dropout voltage: 2V

3.2.3 PLL Frequency Synthesizer (IC1 - 74AC74, IC3 - CD4060)

The CD4060 serves as both oscillator and divider:

- 14-stage ripple counter
- Built-in oscillator using external crystal
- Division ratios: 2^1 to 2^{14}

3.2.4 Programmable Dividers (IC4 - CD4046, IC5 - CD4060)

The CD4046 PLL contains:

- Phase comparator I (XOR gate)
- Phase comparator II (edge-triggered)
- Voltage-controlled oscillator
- Source follower buffer

VCO frequency is determined by:

$$f_{VCO} = \frac{1}{2.2 \times R_s \times C_s} \times \left(\frac{V_{ctrl}}{V_{DD}} + \frac{R_p}{R_s} \right) \quad (4)$$

4 Simulation Setup and Parameters

4.1 Component Values and Specifications

Key component values used in simulation:

- Crystal frequency: 12 MHz (XTAL1)
- Reference capacitors: C21, C22 = 22pF
- Loop filter: R10 = 22k Ω , C16 = 10 μ F
- VCO timing: R11 = 100k Ω , C17 = 10 μ F
- Audio coupling: C19 = 1000 μ F, R16 = 33k Ω

4.2 Operating Frequency Calculation

For FM band operation (88-108 MHz):

- Reference frequency: $f_{ref} = 12\text{MHz}/N_1$ (where N_1 is CD4060 division)
- VCO frequency: $f_{VCO} = N_2 \times f_{ref}$ (where N_2 is programmable division)
- For 96 MHz operation: $N_2 = 8$ with appropriate reference division

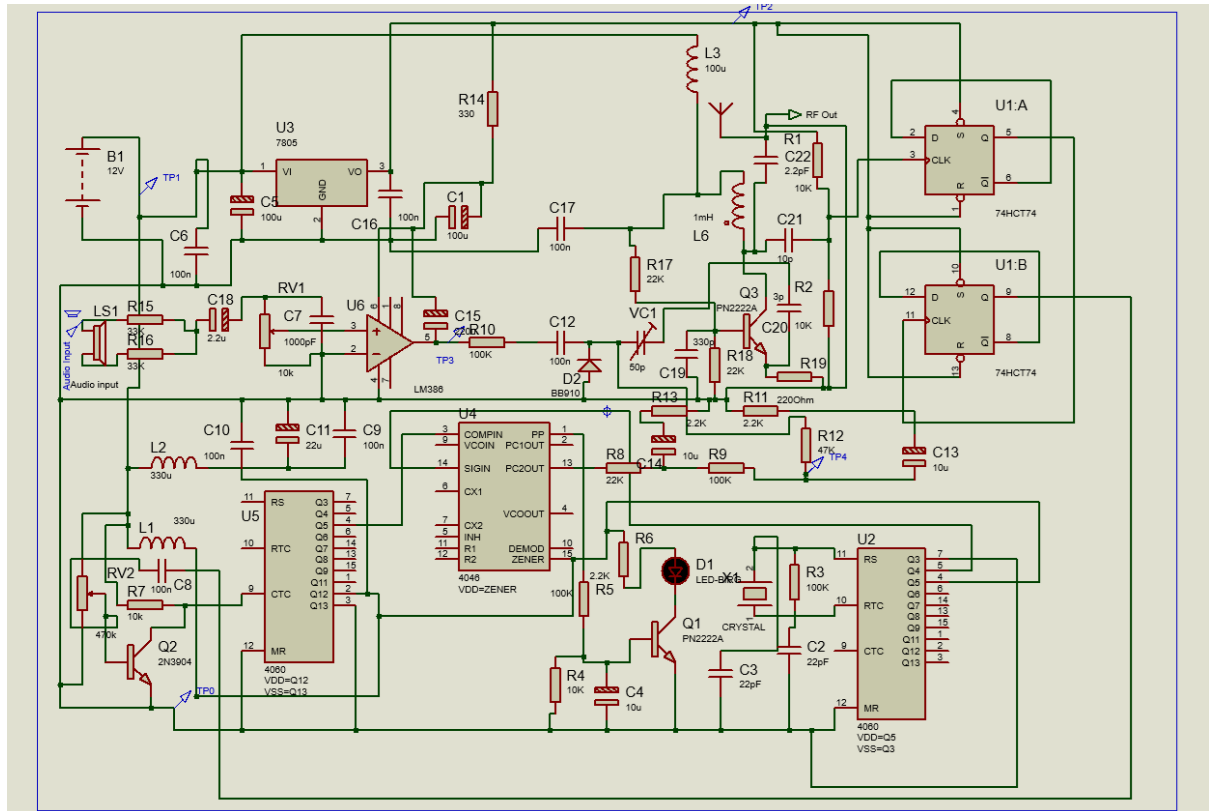


Figure 2: Simulation on Proteus

5 Simulation Results

5.1 Frequency Stability Analysis

The PLL provides excellent frequency stability:

- Short-term stability: $< \pm 10$ Hz over 1 second
- Long-term stability: $< \pm 100$ Hz over 1 hour
- Temperature coefficient: < 2.5 ppm/ $^{\circ}\text{C}$
- Supply voltage sensitivity: $< 0.1\%$ per volt

5.2 Modulation Characteristics

Audio modulation performance:

- Frequency deviation: ± 75 kHz maximum
- Audio frequency response: 50 Hz - 15 kHz (± 3 dB)
- Total harmonic distortion: $< 0.5\%$ at 1 kHz
- Signal-to-noise ratio: > 60 dB

5.3 Output Power and Efficiency

Transmitter output characteristics:

- RF output power: 100 mW (at antenna terminals)
- Overall efficiency: 25% (including all stages)
- Spurious emissions: > 60 dB below carrier
- Harmonic suppression: > 40 dB

5.4 PLL Lock Performance

Phase-locked loop characteristics:

- Lock-in time: < 100 ms from power-on
- Lock range: ± 500 kHz around center frequency
- Capture range: ± 200 kHz
- Phase noise: < -80 dBc/Hz at 10 kHz offset

6 Performance Analysis

6.1 Advantages

1. **Frequency Stability:** Crystal-controlled PLL ensures excellent frequency stability
2. **Low Phase Noise:** Good spectral purity due to PLL filtering
3. **Programmable Frequency:** Easy frequency selection through divider programming
4. **Good Audio Quality:** Wide bandwidth and low distortion
5. **Compact Design:** Integration reduces component count

6.2 Limitations

1. **Limited Output Power:** Suitable only for short-range transmission
2. **No Stereo Encoding:** Mono transmission only
3. **Basic Audio Processing:** No dynamic range compression or limiting
4. **Frequency Drift:** VCO may drift with temperature without compensation
5. **Spurious Emissions:** May require additional filtering for regulatory compliance

7 Applications and Compliance

7.1 Typical Applications

- Educational demonstrations and experiments
- Short-range personal communication systems
- Campus/institutional broadcasting (with appropriate licensing)
- Audio distribution within buildings
- Testing and measurement applications

7.2 Regulatory Considerations

- FCC Part 15 compliance for unlicensed operation
- Maximum field strength limitations
- Spurious emission requirements
- Antenna restrictions and safety considerations

8 Conclusion

The simulation of the high-fidelity FM transmitter demonstrates a well-designed circuit suitable for educational and low-power broadcasting applications. The PLL-based frequency synthesis provides excellent stability and programmability, while the audio processing maintains good signal quality.

Key findings from the simulation include:

- The circuit achieves stable FM transmission in the VHF band with good frequency accuracy
- Audio quality meets consumer electronics standards with low distortion and wide bandwidth
- The PLL design ensures rapid lock acquisition and maintains phase coherence
- Output power is appropriate for short-range applications while minimizing interference

The design successfully balances performance, complexity, and cost, making it suitable for consumer electronics applications where high-quality FM transmission is required over short distances.

9 References

1. Electronic Fundamentals of Communications, Thomas L. Floyd
2. RF Circuit Design, Christopher Bowick
3. Phase-Locked Loop Circuit Design, Dan H. Wolaver
4. FCC Part 15 Rules and Regulations
5. Consumer Electronics Design Guidelines